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Task Description:

The detection of particle radiation induced alterations in the genome in different tissues in vivo is important for the assessment of both short- and long-term risks involved in space explorations. However, such studies are difficult due to (a) the lack of adequate selectable somatic target genes in mammalian systems and (b) the clonogenic potential of the somatic cells chosen. Mutagenic changes that affect the function of non-dividing tissues may be especially deleterious during long-term space flights. The advent of transgenic rodent gene mutation models has provided a means to rapidly assay, in a statistically reliable way, for mutations in the DNA from any tissue of the body over the period of the life span of the animal. These models are the only systems currently available for evaluating both short and long-term tissue specific mutagenic events. They have been accepted by regulatory agencies and are used in genetic toxicology studies on a wide variety of clastogenic and mutagenic agents. In the work proposed here, we have chosen to use a unique plasmid-based lacZ transgenic mouse system to determine the mutagenic potential of charged particle radiations relevant to the space environment. The lacZ transgenic mouse has been constructed so that every cell of the animal contains multiple copies of an integrated target gene. The use of this system allows us to recover and quantify radiation-induced mutations in all tissues of the animal, especially in tissues with limited cell turnover potential, and therefore allows tissue-specific radiation-induced changes to be measured as a function of time post radiation. The nature of these mutations can also be revealed by restriction fragment length polymorphisms (RFLP) and sequence analysis.

We hypothesize that physical parameters of HZE radiation induces tissue-specific responses and aim to use the transgenic animal model in a multi-prong approach to extend our current knowledge to test three specific aims: (1) Examine the genetic damage to the genome by monitoring changes in transgene MF and cytogenetic changes in the peripheral blood and bone marrow lymphocytes as a function of fluence, dose protraction of either pure or mixed particle beams. Beam characteristics in space are critical variables in space and tissue-specific biological effects are important in determining relative risks in space explorations; (2) Measure the dose- and temporal- dependence of HZE radiation-induced alterations in gene expression and the transgene in liver tissue and compare the results with those previously measured in the brain and the spleen. Excess relative risk in liver cancer and non-cancer liver disease or cirrhosis has been demonstrated in Atomic bomb survivors and thorotrast patients; (3) Advance our understanding of the molecular basis for mutation by surveying the transcription activation of target genes in the brain tissue. We will focus our research on examining the expression of proinflammatory cytokines, eg. TNFalpha and interleukin-1, and the neuronal cell adhesion molecule NCAM-1. These genes have been shown to be induced in the mouse brain after low LET exposures.

Knowledge gained in this ground-based work is directed at understanding tissue and organ response to space radiation and will aid both in the estimation of risk to astronaut health and the design of appropriate countermeasures to ameliorate the negative effects in space flights.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The space atmosphere is comprised of particle radiation in a wide spectrum of higly energetic and charged (HZE). Exposures to these HZE radiations poses a serious health hazard to humans in lng-term manned space explorations. We are using the plasmid-based lacZ transgenic mouse mutation model system to examine the acute and long-term tissue specific mutagenic potential after exposure to radiation with different qualities. Knowledge gained in this ground-based work is directed at understanding tissue and organ response to space radiation and will aid both in the estimation of risk to astronaut health and the design of appropriate countermeasures to ameliorate the negative effects in space flights.

During the course of this 4-year program, we have participated in numerous NSRL runs, exposing our lacZ transgenic mice to beams ranging in LET form less than 20 (oxygen) up to greater than 200 keV/µm (iron) and energies of particles ranging from 600 MeV/amu to 1 GeV/amu. A summary of results relevant to each of the original proposed aims is described below:
We have examined the changes in lacZ MF in brain and spleen and the frequency of MN-RET (fMN-RET) in the peripheral blood after low, protracted exposures of iron ions or proton with the same high 1 GeV/u energy. Peripheral blood was harvested from animals exposed to 5 daily low dose (2 – 20 cGy) fractions of 1GeV/u Iron ions to simulate protracted radiation exposures in the space environment. Our results (published in Advances in Space Research, 2007) show that the level of %MN-RET at 24, 48 and 72h after the protracted dose exposures are lower than if the doses were administered in one bolus.

To evaluate whether the ‘sparing’ effects seen after fractionated iron ions are dependant on the LET of the incident radiation, we conducted a parallel study exposing animals to the same dose-range and single or fractionation scheme with 1 GeV/u protons. We noted a dramatic drop in the fre¬quency of total circulating reticulocytes (fRET) shortly after a single dose of proton irradiation, a definitive indicator that particle radiation induced an overall dose-dependent systemic toxicity in the hematopoietic system. This was followed by complete recovery of RET counts within 1 week after exposure. The dose-dependent increase in fMN-RET was observed within 24 h after irradiation. fMN-RET peaked at 48 hr after the acute exposure, but returned to normal levels by 1 wk. No significant increases in the fMN-RET were observed in the samples obtained from any of the animals exposed to 10 – 100 cGy of total proton dose at any time after fractionated exposures. Unlike the dramatic drops in total RET counts seen after acute exposures, samples obtained from animals exposed to fractionated doses were only mildly affected, another indication that an active and rapid recovery from radiation damage is in place in this in vivo model system.

In order to gain insight into the dynamic processing of lesions produced by the particle tracks, lacZ mutants harvested from single- or fractionation- irradiated animals at 8-wk after radiation were characterized using the restriction fragment length polymorphism (RFLP) patterns. Although it appeared that the dosing regime did not results in statistically significant differences in the total MF after 1 Gy iron ions in brain tissue, the % of size change (deletion) mutants was 2.5-fold higher (16%) in samples obtained from single-dosed animals compared to the fractionated animals (6%). On the other hand, even though the total MF after 2 Gy iron ions in brain samples was significantly higher if the dose was administered in a single dose instead of 5 equal daily fractions, the % of point, deletion or insertion mutants were comparable.

The average MF in spleen after 1 or 2 Gy fractionated iron-treated animals was comparable to the spontaneous levels and significantly lower than that obtained after the single dose treatment. Characterization of the mutant spectrum revealed that, unlike those seen in the 2 Gy single dose treatment group, most of the mutants that contribute to the MF are point or deletion mutants with no recombination mutants. Our current results suggest that the lowered MF after

Task Progress:	repeated daily dosing of iron ions may be due to cell attrition in the spleen tissue. This is in contrast to the tissue response when 2 Gy was administered in a bolus where cells appear to undergo repair through recombinational pathways. Additional information at early times after radiation exposure will provide insight in the initial genomic lesions and provide information on the dynamic processing of particle radiation induced tissue damage. We have made significant progress in evaluating the effect of beam quality of particle radiation using the transgenic mouse mutagenesis model system. Animals were exposed to beams with LETs ranging from <1 to ~200 keV/µm. Analysis of the profile of changes in fMN-RET harvested at 48 hr after exposure as a function of incident particle fluence showed that: (1) there is a progress shift of the MN-RET response curves to right with increasing fluence, (2) particle beams with similar LETs, such as 290 MeV/u Carbon ions and 1 GeV/u Oxygen ions, induce similar levels of increase in fMN-RET after irradiation, and (3) the appearance of “hook” structures in each of the curves suggests that, for each beam-energy, a maximum saturation effect can be attained. Additional careful consideration of particle fragmentation as the different beams traverse through the whole animal is needed before any firm conclusions can be drawn with these results. Genomic tissue damage, expressed as mutations in the reporter transgene in the brain and spleen tissues harvested at 8 weeks after 1 GeV/n titanium and oxygen ion irradiation showed an increase in MF of up to 2-fold above spontaneous levels as a function of dose after titanium ion exposure. Comparison of these results with MF yields after 1 GeV/u iron ions and 250 MeV/u protons suggests that titanium and oxygen beams are less mutagenic than iron or proton ions. The assay system used in this project provides a glimpse of the residual damage in surviving cells at a point in time. We postulate that the lower level of MF yield after titanium and oxygen ions could be due to the increased effectiveness of titanium or oxygen beam in cell killing, resulting in a lower number of surviving cells carrying genomic damage. Changes in MF in the two tissues as a function of incident fluence of particle radiation clearly showed that the mutagenic potential of HZE particle beams is dependent on the particle fluences. The neuronal cell adhesion molecule (NCAM) is known to be associated with neuronal cell plasticity, synaptic associations as well as the regulation of higher cognitive function. We compared the relative expression of NCAM transcripts in brain tissue at 1 week after a range of 1 GeV/u Iron ions and 250 MeV/u protons. Our results show a continuous decline in NCAM transcription with increasing Iron dose, suggesting potential major alteration in cellular communication and remodeling of brain tissue. On the other hand, NCAM expression in brain tissues after protons increases relative to the control levels as a function of dose after protons. These results suggest that beam quality impacts the expression of adhesion molecule expression. TNF alpha, a proinflammatory cytokine, is known to play a role in the late pathogenesis of the irradiated brain. TNFa has been specifically associated with aberrant cellular responses such as astrocyctosis, decreased expression of oligodendrocytes and myelin-associated molecular markers observed in mouse models, suggesting that cytokine-driven cellular responses. Results from our quantitative evaluation of expression levels showed that TNFa was down regulated at doses as low as 10cGy of iron ions but up regulated at 2 Gy of protons. These results points to the importance of beam quality, as well as dose levels in the regulation of brain tissue-specific responses.
Bibliography Type:	Description: (Last Updated: 04/26/2023)
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